Essential Competencies for Highly Effective 21C/ Beyond 2020 Engineers

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Advancing Computing as a Science & Profession

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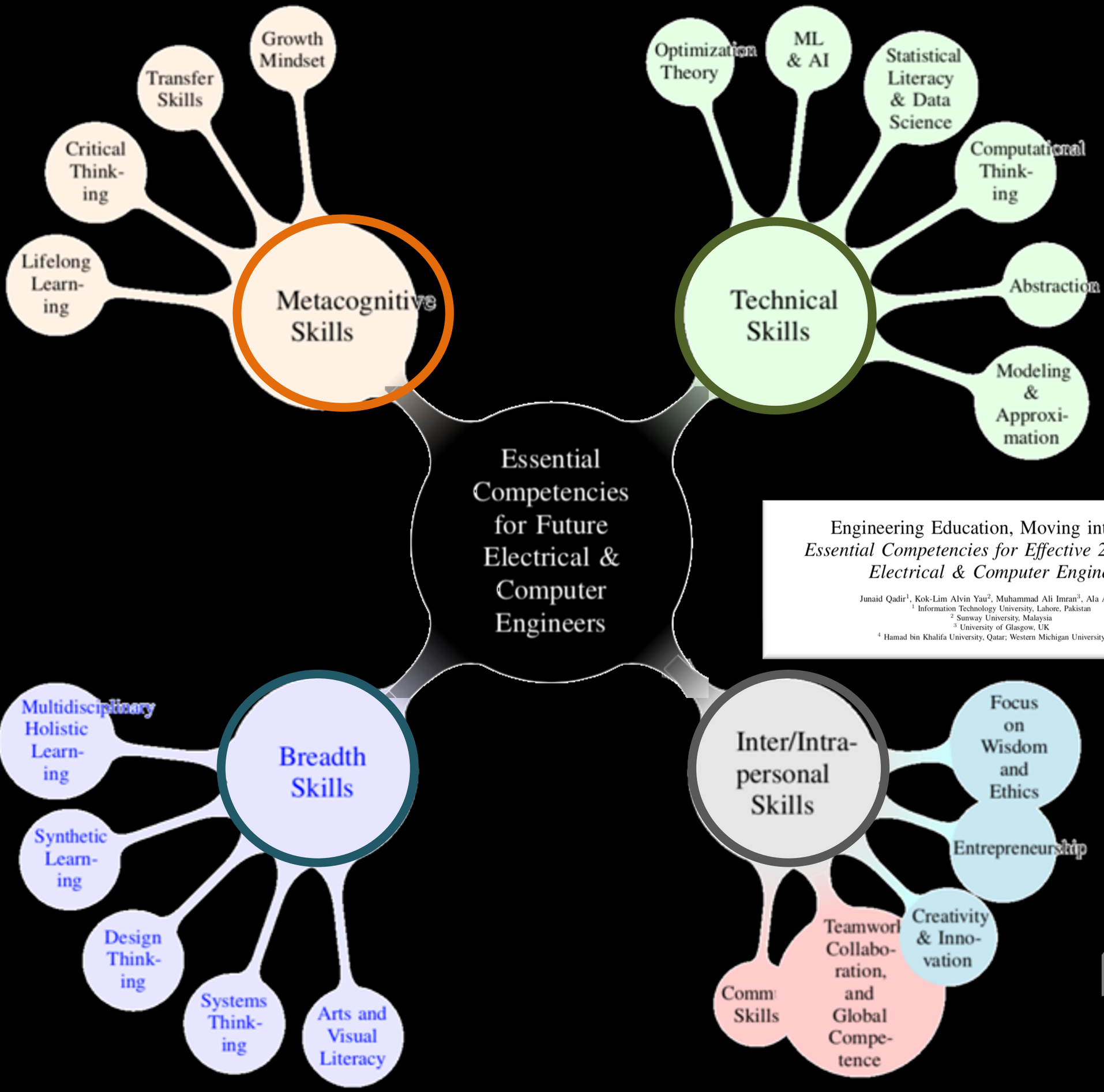

Engineering Education, Moving into 2020s:
Essential Competencies for Effective 21st Century Electrical & Computer Engineers

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4

Some interaction with the audience

1

3

Growth Mindset

Transfer Skills

Critical Think-ing

Lifelong Learn-ing

Optimization Theory

ML & AI

Statistical Literacy & Data Science

Computational Think-ing

Abstraction

Modeling & Approxi-mation

Technical Skills

Essential Competencies for Future Electrical & Computer Engineers

Breadth Skills

Multidisciplinary Holistic Learn-ing

Synthetic Learn-ing

Design Think-ing

Systems Think-ing

Arts and Visual Literacy

Inter/Intra-personal Skills

Focus on Wisdom and Ethics

Entrepreneurship

Creativity & Inno-vation

Teamwork Collabo-ration, and Global Compe-tence

Comm Skills

2

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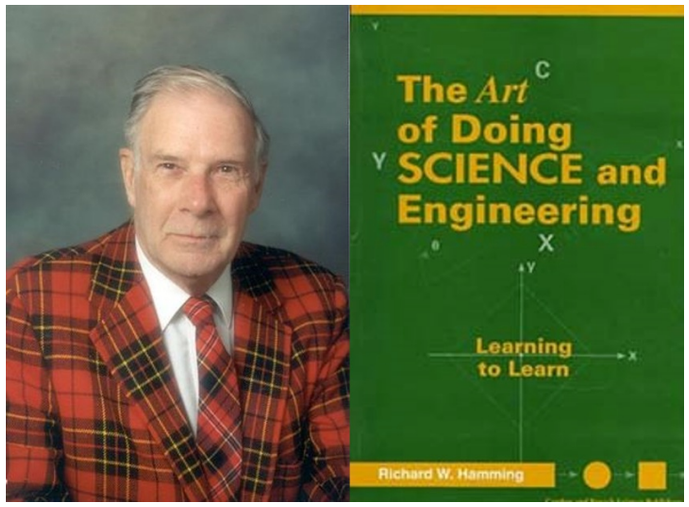
Motivating this lecture: Finding your why?



"No vision,
not much of a future."

"I am preaching the message that, with apparently only one life to live on this earth, you ought to try to make significant contributions to humanity rather than just get along through life comfortably—that the life of trying to achieve excellence in some area is in itself a worthy goal for your life.

It has often been observed the true gain is in the struggle and not in the achievement—a life without a struggle on your part to make yourself excellent is hardly a life worth living."

Lifelong learning

"Teach a man to fish, and you feed him for a lifetime."

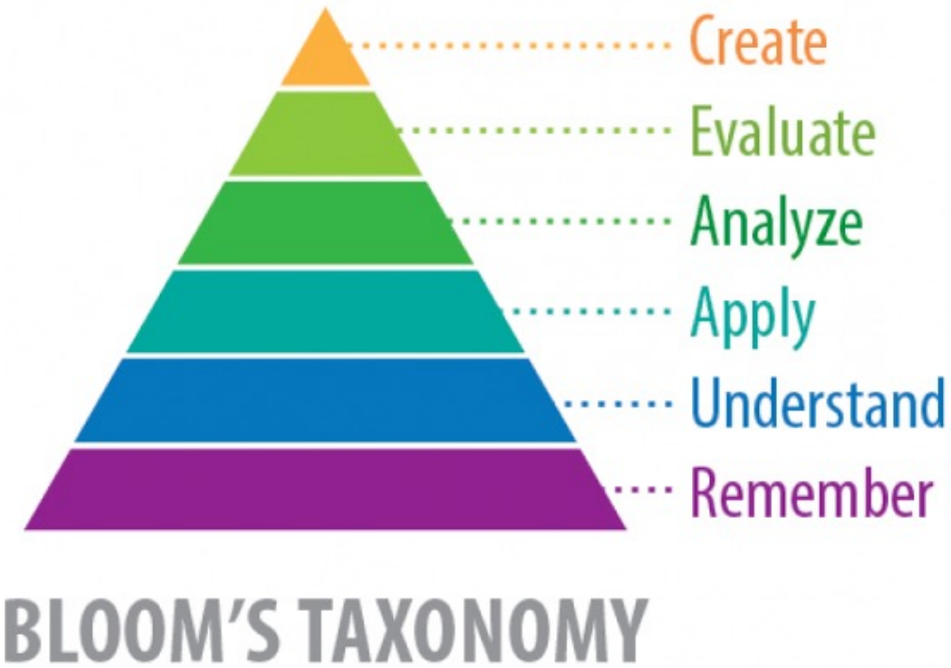
Matthew Effect in Education:

When students acquire meta-skills such as learning how to learn, they bring this skill to every course and reap its exponential benefits

Learning that Transfers

"A primary goal of education is the development and deepening of student understanding of important ideas and processes within, and across, disciplines so that **they can transfer their earning to new situations.**"

—Wiggins and McTighe,
"Understanding by Design"
(UbD framework)



Problem of Overfitting
in machine learning
(failure to generalize)

IMPROVING ONE'S METACOGNITIVE SKILLS

Avoid common mistakes

Learn about most effective techniques

Dollars Versus Diplomas

What every student should know: Seven learning impediments and their remedies

Junaid Qadir

Many well-intentioned students fail to thrive in a university environment, not due to lack of effort, but because of a deficient mind-set and/or methodology. While a wide variety of problems can result in suboptimal learning performance, there are seven main categories of learning impediments that hinder student learning with alarming regularity. You may be facing one, or more, of these impediments. Not to fret though; each of these debilitating impediments can be surmounted with self-awareness and some effort, paving the way for enjoyable, and productive, learning experiences. In this article, I will identify the offending learning impediments and propose plausible remedial policies to counter each.

Is there anyone so wise as to learn by the experience of others?

—Voltaire

First impediment: Having a fixed mind-set

Psychologists have highlighted the self-enforcing propensity of the human mind-set in that human optimism and pessimism are both self-fulfilling. As Henry Ford said, "Whether you think you can or think you can't—you are right." A *fixed mind-set*, a mind-set model proposed by researcher Carol Dweck, takes the pessimistic approach to the concept of intelligence and skill development. It characterizes a belief that intelligence or talent are *fixed* human traits that cannot be significantly developed beyond a certain degree. Students with fixed mind-sets obsess over documenting their intelligence instead of focusing on developing their intelligence. Many fixed mind-set students are gifted students who have not learned how to struggle or the value of hard work. While they aced school subjects with little effort, when confronted with more significant intellectual challenges in a university setting that require more effort, these students capitulate easily while resorting to the excuse of a lack of aptitude. Why else would they—being such intelligent students—be confused?

What can be done: The antidote to this problem is to: 1) have a growth mind-set, 2) aim for mastery, and 3) use intrinsic motivation.

Catch phrase for the solutions: The sky is not the limit—your mind is. Conquer your mind to realize your potential.

Remedy 1: Have a growth mind-set

A *growth mind-set* student believes that hard work, grit, and

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Careers and Grad Ed

Learning 101: The untaught basics

Junaid Qadir and Muhammad Ali Imran

Despite the accessibility of a wealth of information in the current era—books, universities, and online massive open online courses (MOOCs)—well-intentioned and hard-working students often fail to learn effectively due to deficient learning techniques or improper mind-sets. Two things, in particular, hinder students from achieving their potential. First, the students' intuition regarding how learning works is often flawed and counterproductive; second, despite significant progress in the research discipline of "learning sciences," these hard-earned scientific insights have not yet filtered their way through the research community to the students who stand to benefit most from this knowledge.

In this article, we aim to popularize the important insights from learning-science researchers by making it accessible to students, who continue to largely use suboptimal intuitive learning techniques (not knowing any better). The benefits of becoming wise to these insights are immediate and substantial: we can use this knowledge to understand and calibrate our learning, and it can also facilitate efficient learning.

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Common learning impediments

To learn optimally, we need to discover how to learn effectively and also to identify and avoid common learning impediments. In our previous work, we identified seven common learning impediments (shown in Table 1). These impediments emerge from deficient mind-set (numbers 1 and 4), focus (2 and 3), support (5 and 6), and metacognition (7). These issues are not insurmountable; three potential remedies for each learning impediment are identified in our previous work as shown in Table 1. A major impediment—arguably the mother impediment of all the listed impediments, if considered broadly—is number 7, "Not learning how to learn." The overarching remedy for this problem is to spend time developing better metacognitive skills. Our aim is to succinctly present the relevant research findings in an accessible manner so that you can use this knowledge to improve your metacognitive skills and learning strategies.

Learning how to learn

The most useful piece of learning for the uses of life is to unlearn what is untrue.

—Antisthenes

Despite our intimacy with the task of learning through the trials and errors of everyday living), research has shown that our metacognitive judgment about learning (i.e., our intuitive understanding of how we learn) is flawed, and many of our well-established and commonly accepted intuitive notions are not only incognizant of scientific findings about learning but oppose the optimal course of action. Unfortunately, the common societal attitude, and the general assumption in colleges and universities, is that students can automatically figure out the art and science of learning by themselves. This is reflected in the way we emphasize the teaching of content and subjects in educational institutions but not the dissemination of information regarding an optimal learning strategy.

There is a Taoist saying that states, "To gain knowledge, one should add something every day, but to gain wisdom, one should subtract something every day." We propose that students should supplant the erroneous intuitive learning model with an improved metacognitive model of how learning

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Via Negativa

Dollars Versus Diplomas

What every student should know: Seven learning impediments and their remedies

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Many well-intentioned students fail to excel in university environment, not due to lack of effort, but because of a deficient mindset and/or methodology. While a wide variety of impediments can retard optimal learning performance, there are seven main categories of learning impediments that hinder student learning with alarming regularity. You may be facing one, or more, of these impediments. The first (tough) study of these debilitating impediments can be surmounted with self-awareness and some effort, paving the way for enjoyable, and productive, learning experiences. In this article, I will identify the offending learning impediments and propose plausible remedial policies to combat them.

> In three degrees we arise on
> to learn by the experience of
> others?
>
> —Voltaire

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What can be done? The antidote to this problem is to: 1) have a growth mind-set, 2) aim for mastery, and 3) use intrinsic motivation.

Let us now 'do' for the solution. The sky is not the limit—*your* mind is. Conquer your mind to realize your potential.

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J. Qadir,
"*What every student should know: Seven learning impediments and their remedies*,"
IEEE Potentials, vol. 34, no. 3, pp. 30–35, 2015.

Impediment 1) Having a Fixed Mindset

Solution: Conquer your mind to realize your potential.

Impediment 2) the Failure to Engage 'Yourself' in Learning

Solution: Knowledge will not give a part of itself until you give to it all of yourself.

Impediment 3) the Failure to Manage Time

Solution: Until we can manage time, we can manage nothing else.

Impediment 4) Failing to Realize that Failing is Key

Solution: Err unabashedly and learn—like a child does.

Impediment 5) Failing to Realize that Learning is Social

Solution: No one can do it for you, but you can't do it alone.

Impediment 6) Being a Learning Monogamist

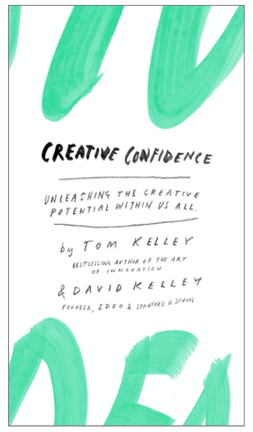
Solution: Encourage learning polygamy.

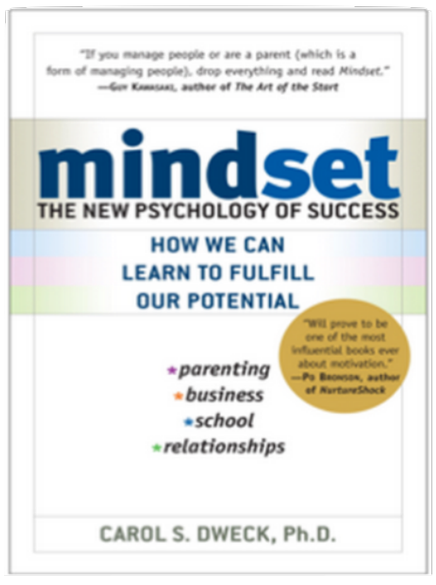
Impediment 7) Not Learning How to Learn

Solution: Don't be an highly qualified A grade sheep.

Issues of Mindset

"the surprising, compelling mathematics of innovation: if you want more success, you have to be prepared to shrug off more failure."

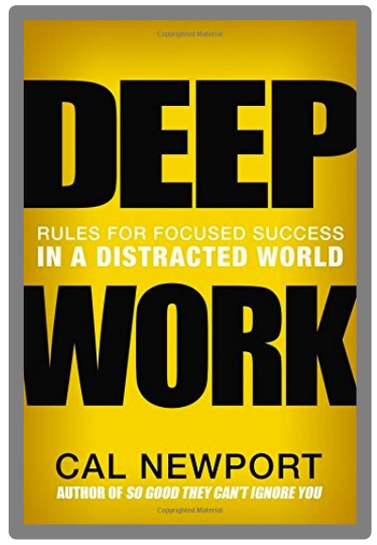




گرتے ہیں شہ سوار ہی میدان جنگ میں
وہ طفل کیا گرے کہ جو گھٹنوں کے بل چلے

Managing the Tsunami of Distractions

High-Quality Work Produced = (**Time Spent**) x (**Intensity of Focus**)



Deep Work Hypothesis:
the few who cultivate this skill, and make it the core of their working, will thrive.

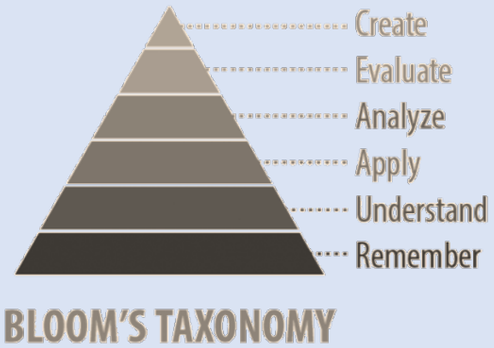
Shallow Work

Noncognitively demanding, logistical-style tasks, often performed while distracted.
***Shallow work**, is low value and easily replicable.*

Deep Work

Professional activities performed in a state of distraction-free concentration that push your cognitive capabilities to their limit.
***Deep work** create new value, improve your skill, and are hard to replicate.*

Questions/ Discussion Mini-Break



BLOOM'S TAXONOMY

Dollars Versus Diplomas

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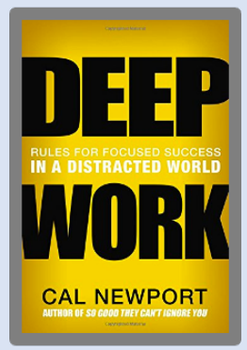

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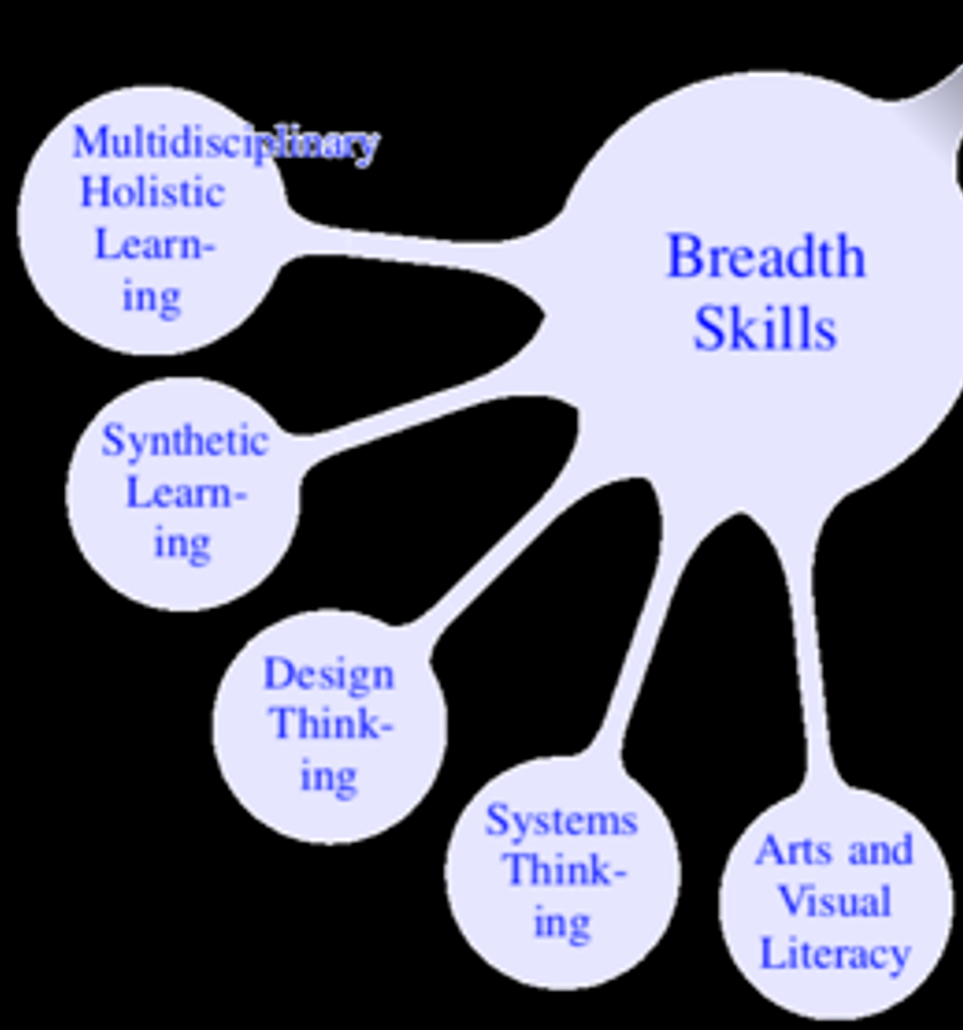
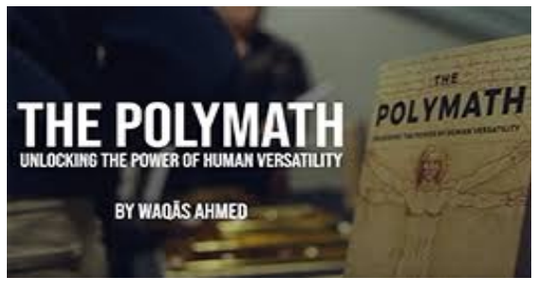
Essential Competencies for Future Electrical & Computer Engineers



Becoming a polymath (multidisciplinary learning)

No one model alone is entirely perfect.

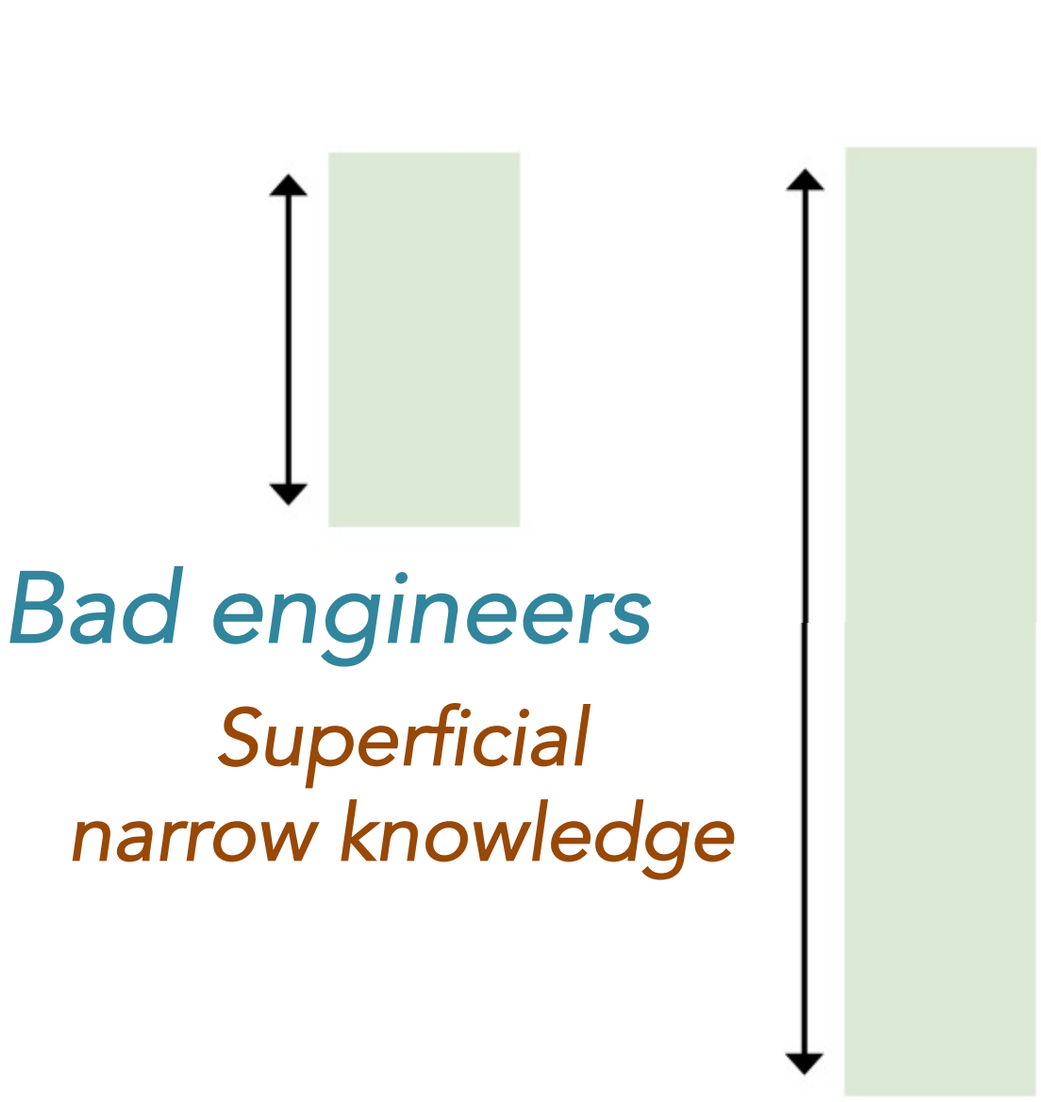
If a hammer is all you have, everything appears like a nail



Combine the arts and the sciences (e.g., for visualizing and generating ideas).

Breath Not At The Cost of Depth

(i shaped vs. I-shaped vs. T-shaped)

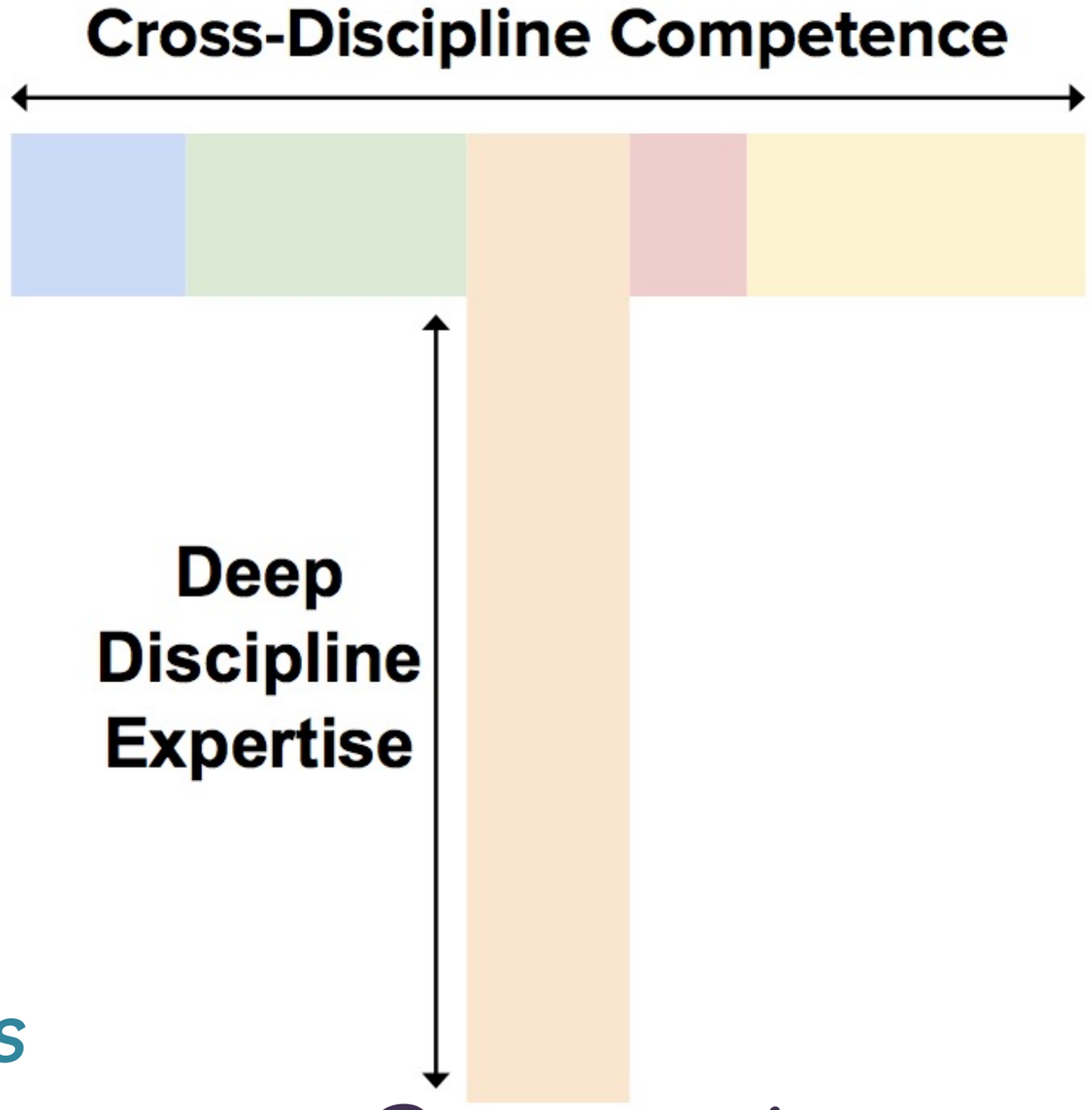


Bad engineers

Superficial
narrow knowledge

Good engineers
Bad problem solvers

Deep (but narrow) expertise



Great engineers
Good problem solvers

Focusing on the whole (systems thinking)

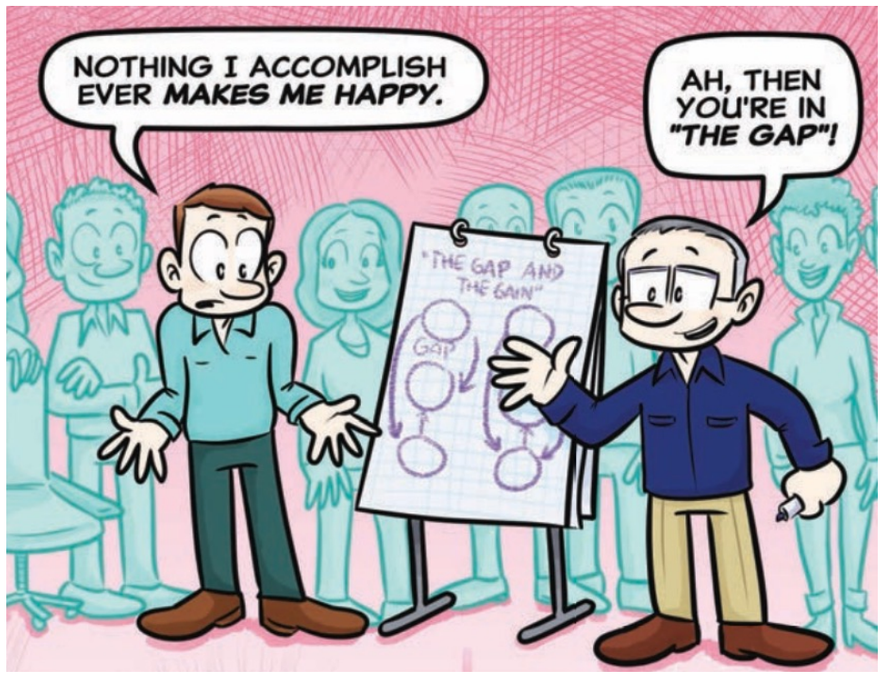
Optimizing the parts vs.
focusing holistically on the interconnections

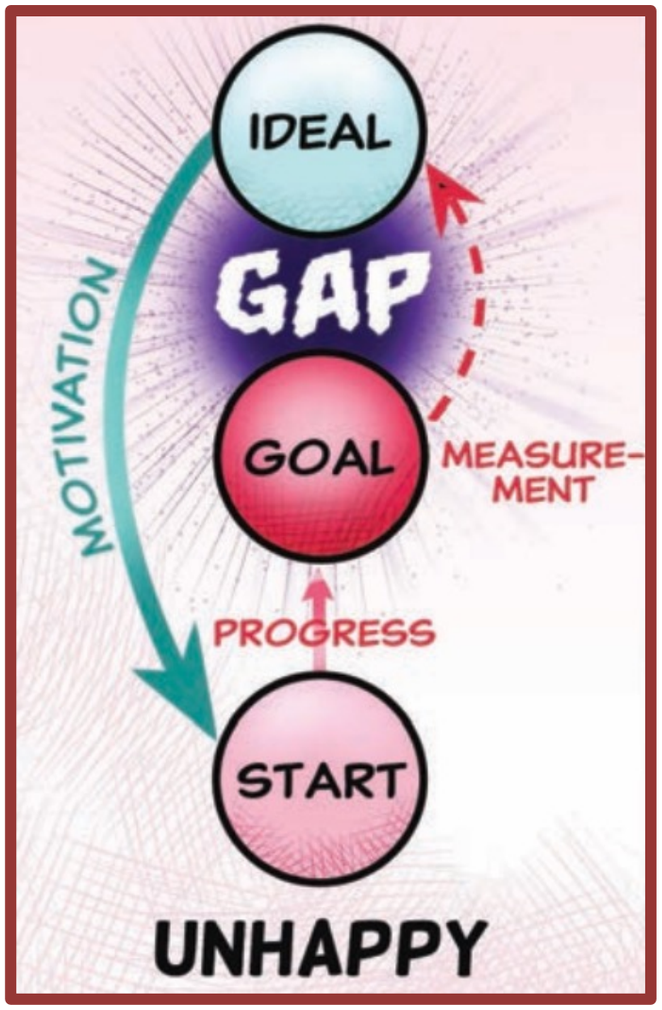
Figure Credit: Naveed Sherwani (CEO, Si-Five Inc.)

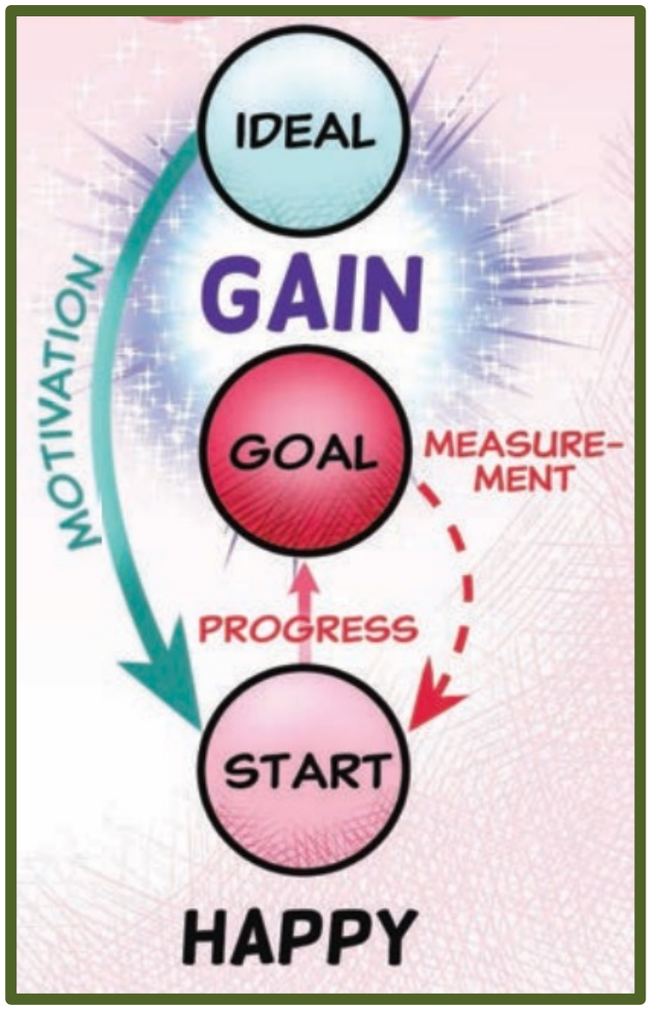
How to Measure: The **Gap** and the **Gain**

ہر اک مقام سے آگے مقام ہے تیرا
حیات ذوقِ سفر کے سوا کچھ اور نہیں

رنگ ہو یا خِشت و سنگ، چنگ ہو یا حرف و صوت
معجزۂ فن کی ہے خُونِ جگر سے نمود

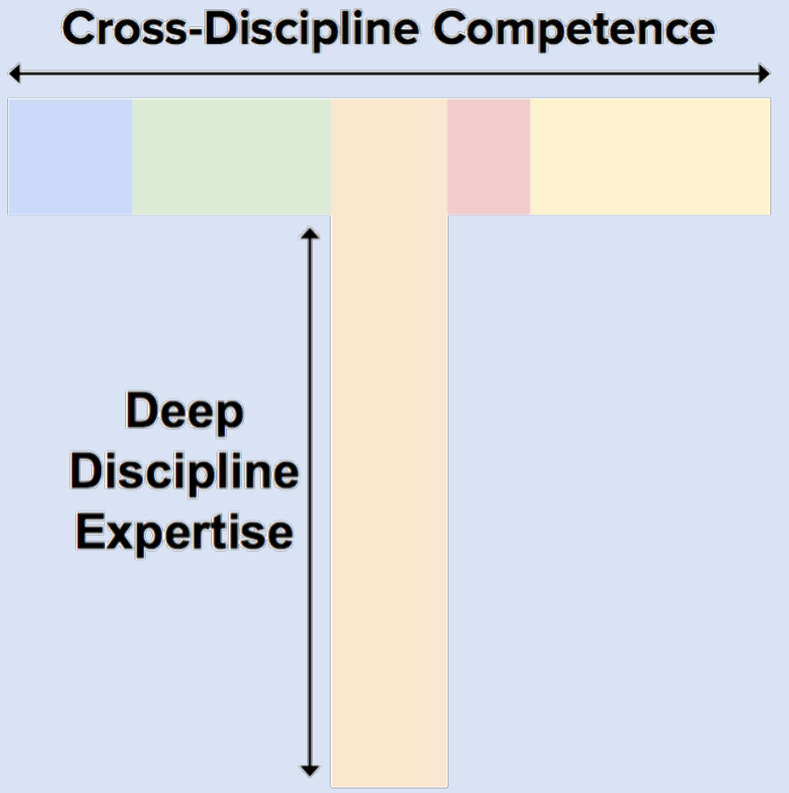


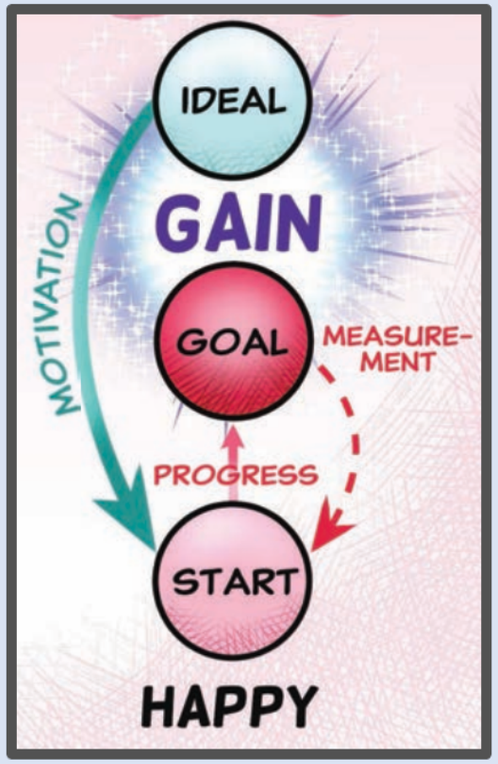


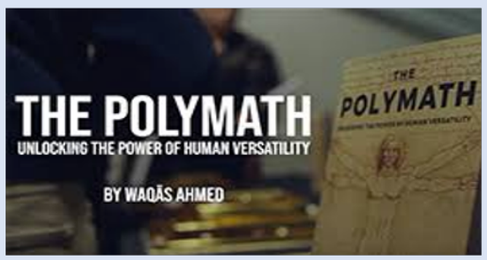


Dan Sullivan, Gap and the Gain

Questions/ Discussion Mini-Break







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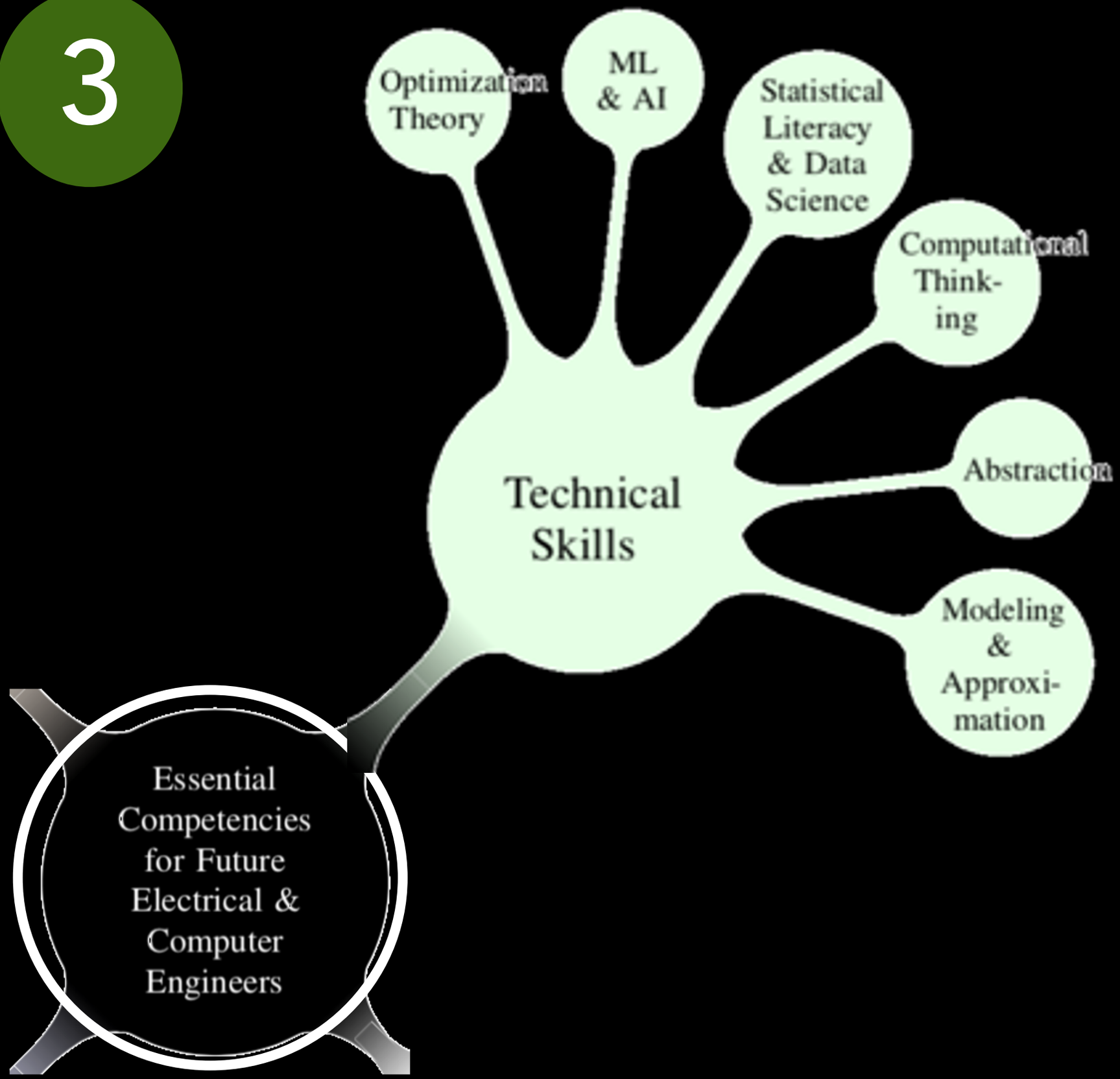
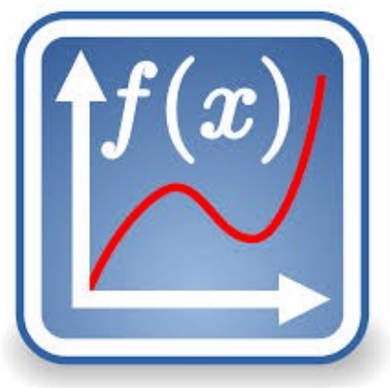

Optimization Theory
ML & AI
Statistical Literacy & Data Science
Computational Think-ing
Abstraction
Modeling & Approxi-mation
Technical Skills
Essential Competencies for Future Electrical & Computer Engineers

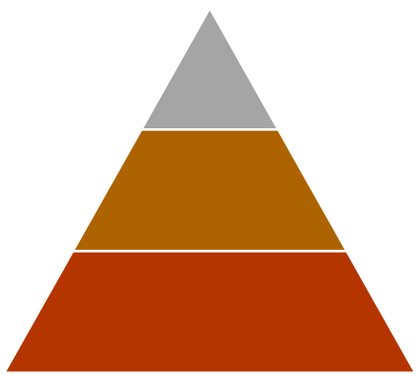
Salient Technical Know-How

Modeling and Approximation



Computational Thinking/Skills

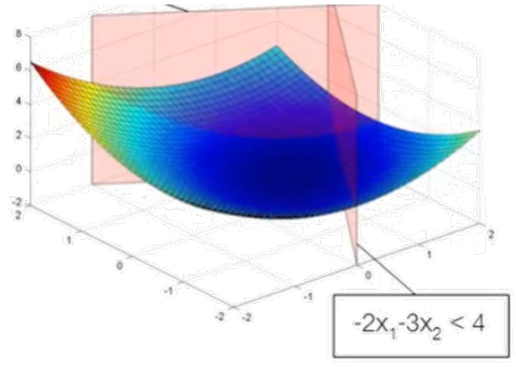
Abstraction



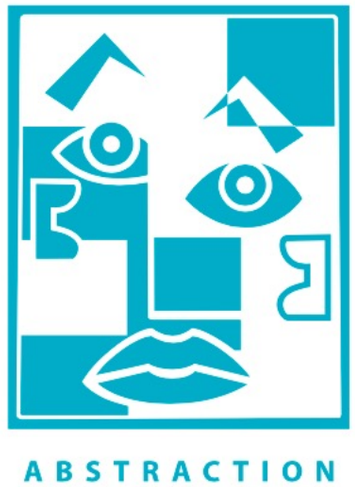
Statistical/Data Literacy

Machine Learning & Artificial Intelligence

Optimization Theory



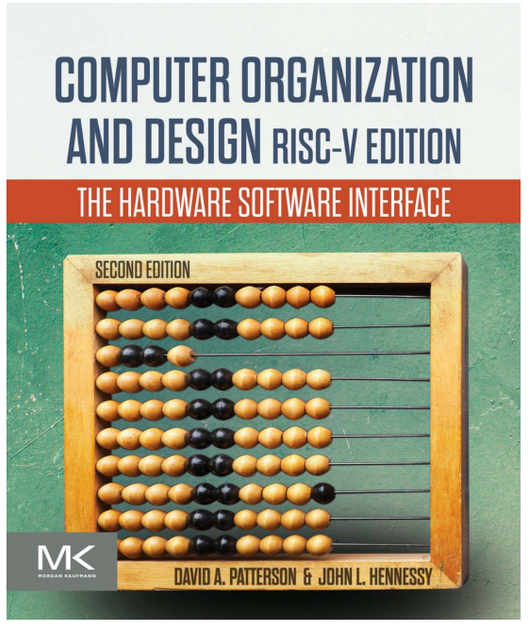
What are the big ideas in your field?

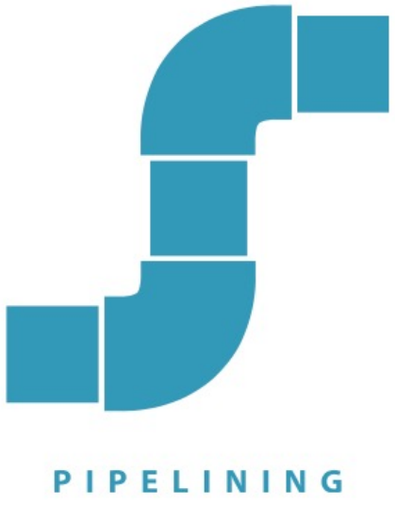


Use Abstraction to Simplify Design

Make the Common Case Fast

Performance via Parallelism





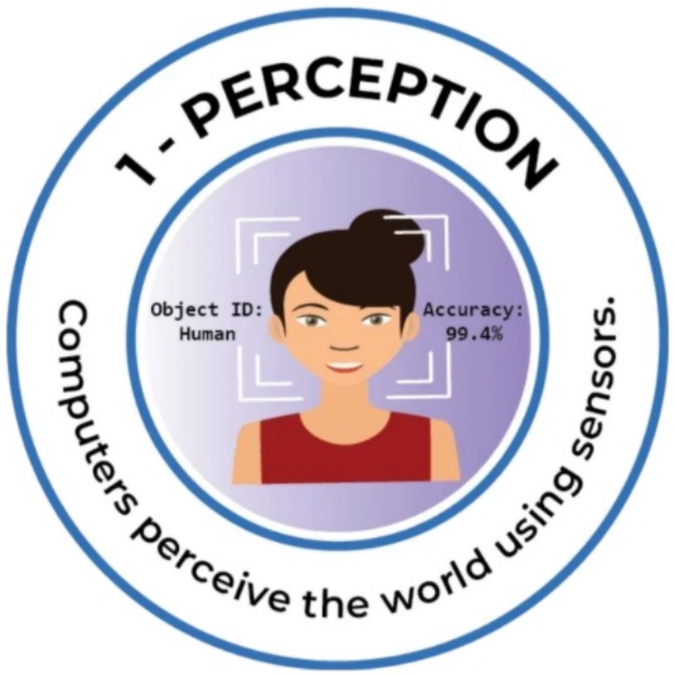
Performance via Pipelining

Performance via Prediction

Hierarchy of Memories

Dependability via Redundancy

Big Ideas in Artificial Intelligence



Big Idea 1 in AI

Big Idea 2 in AI

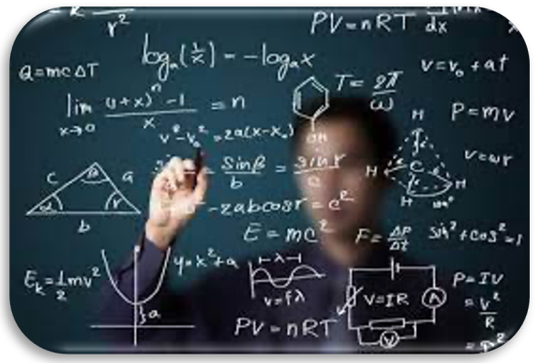
Big Idea 3 in AI

Big Idea 4 in AI

Big Idea 5 in AI

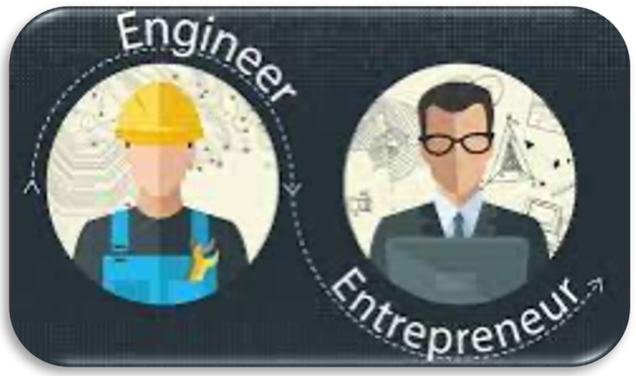
Effective engineers desiderata

Mathematics
Mathematical Modeling
Mathematical Analysis

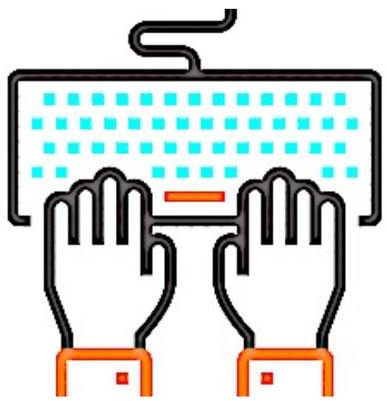


Coding
Programming
Simulation

Engineering
Entrepreneurship
Resourcefulness



Engineering
is a hands on
discipline

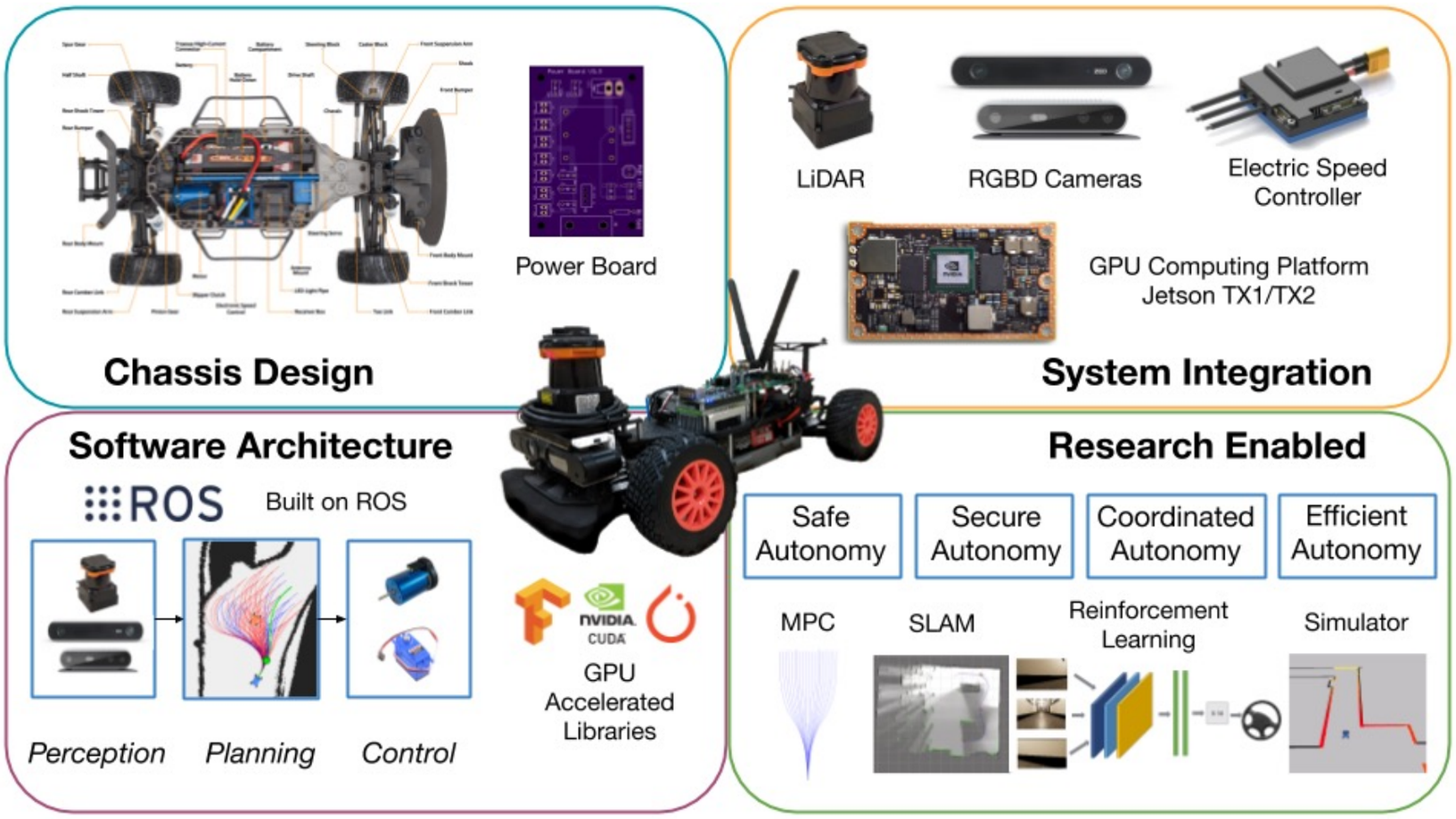


“Hands-on”
Tinkerer
"Hacker Mindset”
“Design Thinking”

AUTHENTICITY
IN EDUCATION

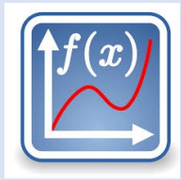
Authentic performance tasks provide opportunities for students to apply their learning in relevant contexts and enable the integration of 21st Century Skills with academic content.

GitHub



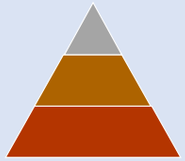
Questions/ Discussion Mini-Break

Modeling and Approximation



Computational Thinking/Skills

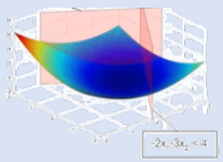
Abstraction



Statistical/Data Literacy

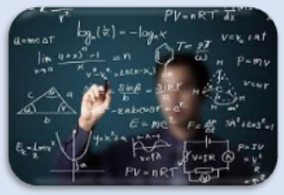
Machine Learning & Artificial Intelligence

Optimization Theory



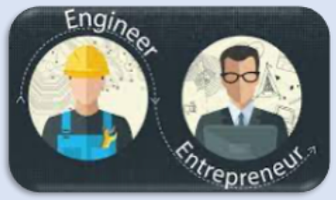
Learning the ***big ideas*** *in your field*

Mathematics
Mathematical Modeling
Mathematical Analysis



Coding
Programming
Simulation

Engineering
Entrepreneurship
Resourcefulness



Engineering
is a hands on discipline

"Hands-on"
Tinkerer
"Hacker Mindset"
"Design Thinking"

Essential Competencies for Future Electrical & Computer Engineers

4

Inter/Intra-personal Skills

Focus on Wisdom and Ethics

Entrepreneurship

Creativity & Innovation

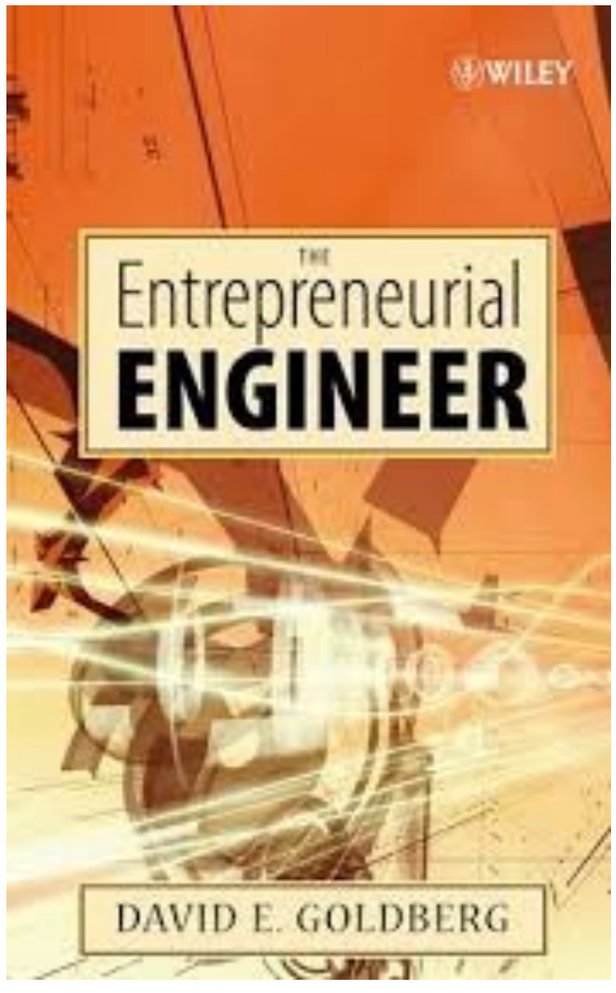
Teamwork Collaboration, and Global Competence

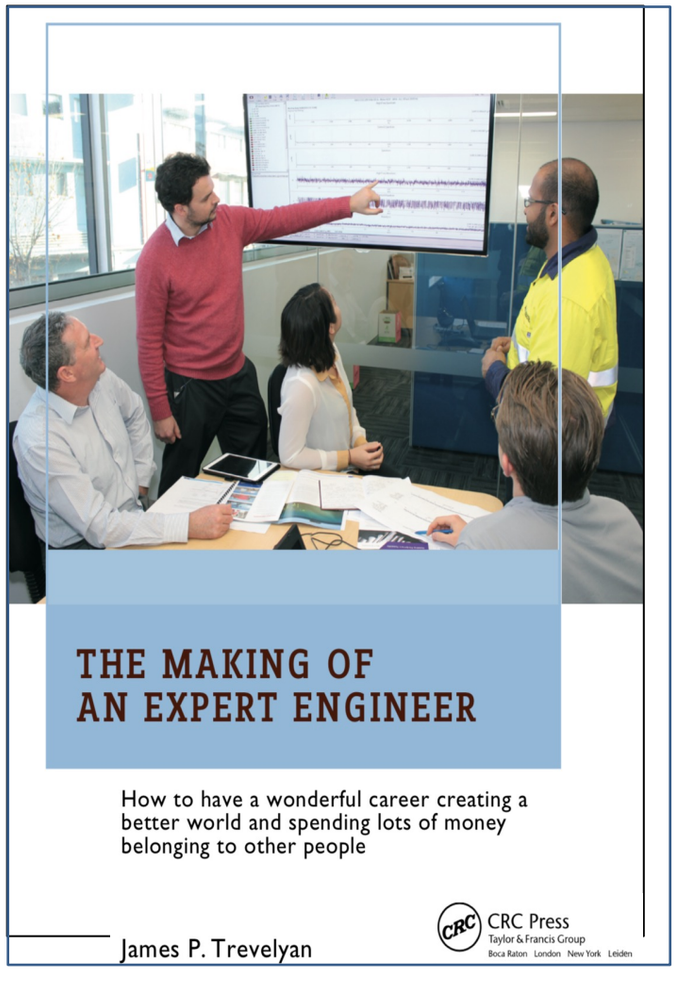
Comm Skills

Entrepreneurship

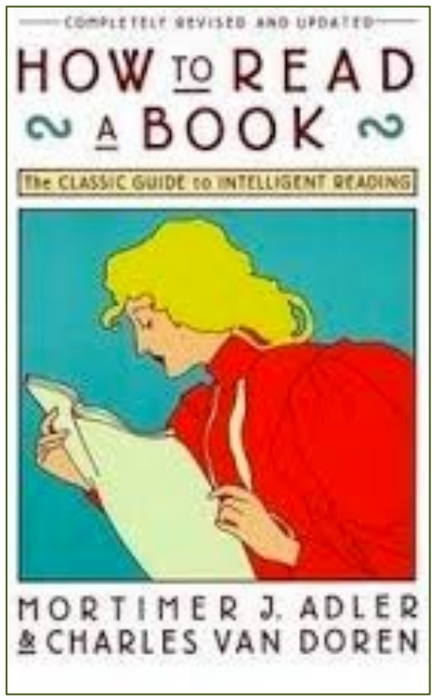
4C Skills for the 21C.

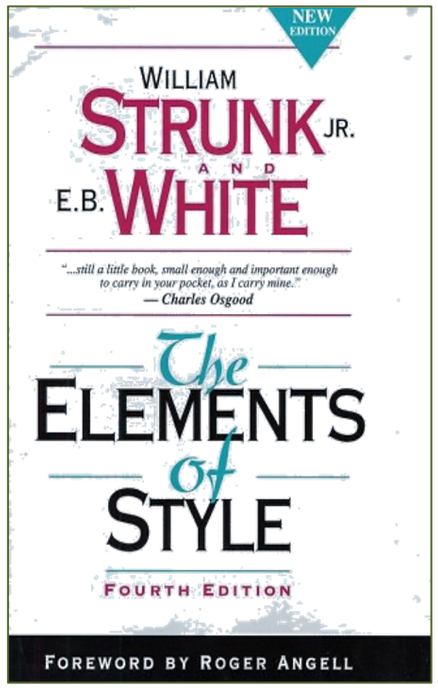
Critical thinking; **C**ommunication; **C**ollaboration; and **C**reativity.

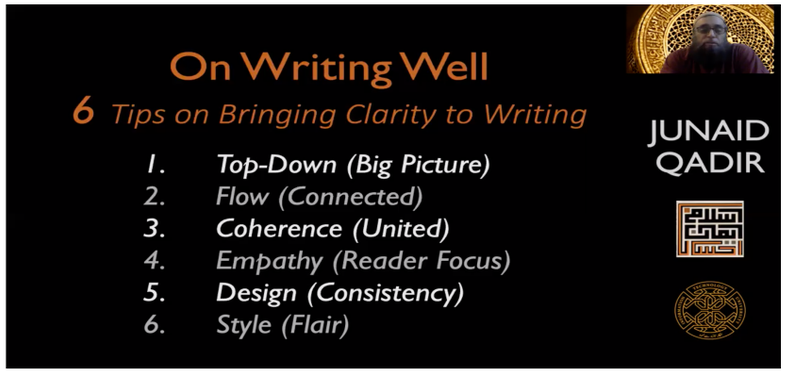




Communication Skills







Writing Course - June 2020

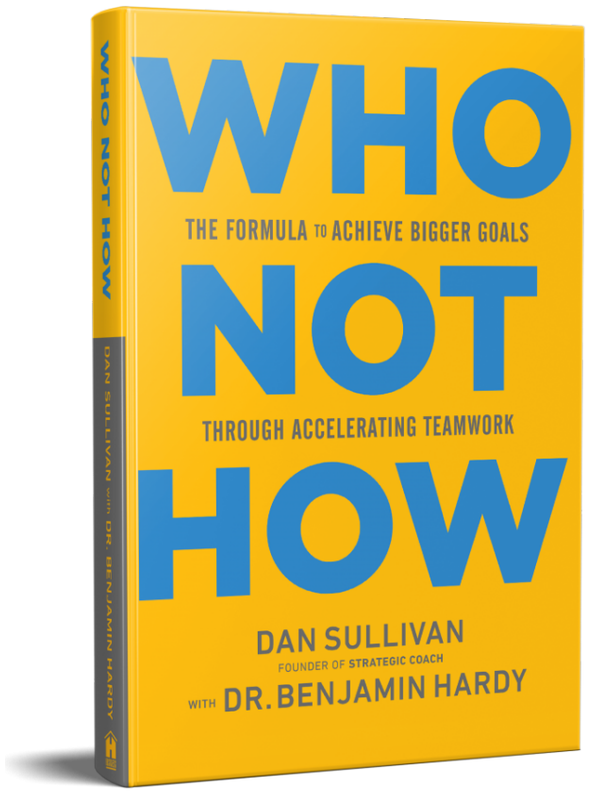
Technical Writing Short Course 6 Tips on Bringing Clarity

https://www.youtube.com/junaidq

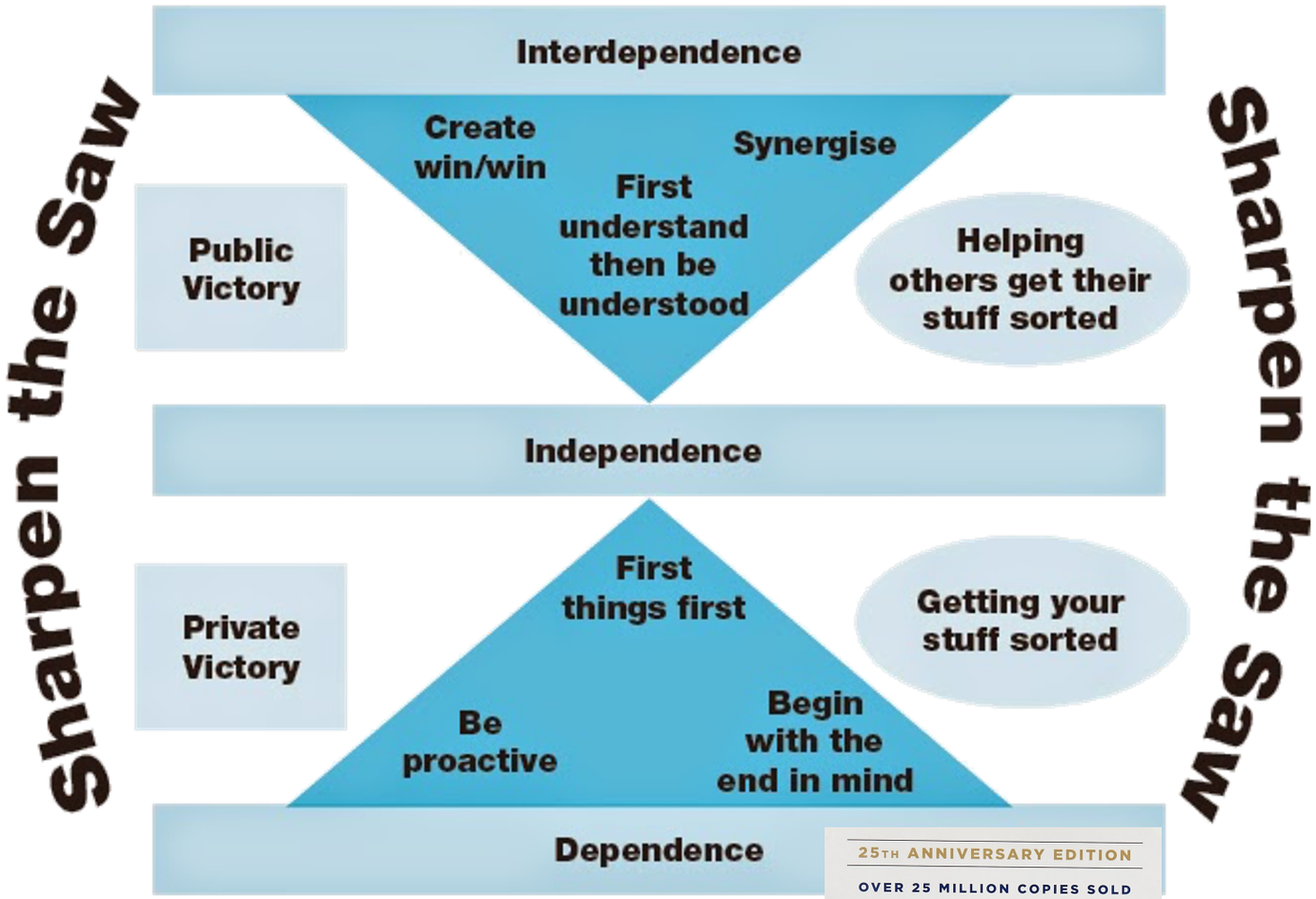
"It is nearly impossible to overstate the benefits of being able to write well. The importance of the written word in storing, sharing, and communicating ideas at all levels of all organizations makes a poor facility with the mechanics of writing a severely career-limiting fault."

John E. West, *The Only Trait of a Leader: A Field Guide to Success for New Engineers, Scientists, and Technologists*, 2008

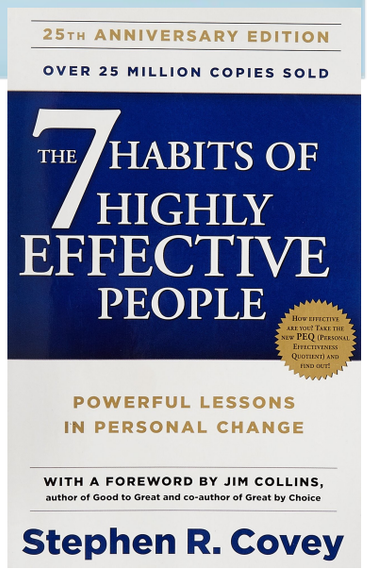
Team Work and Collaboration Skills



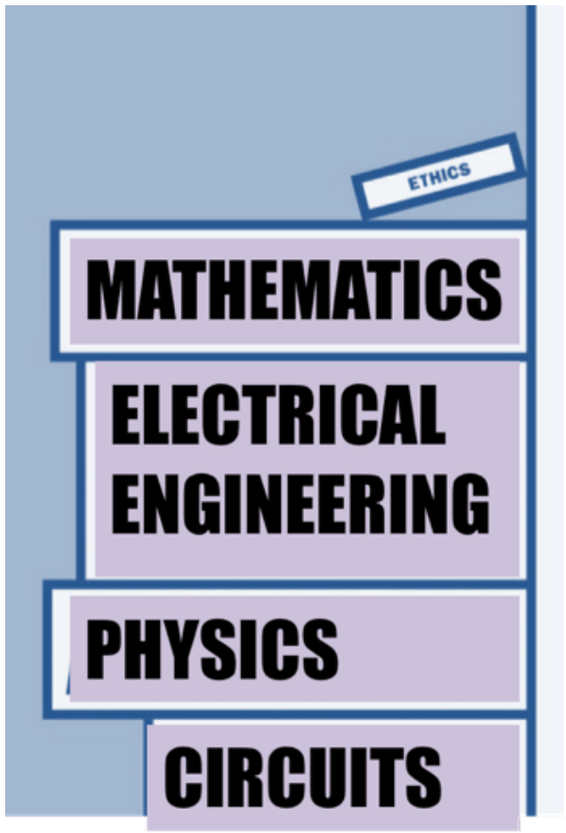
You cannot achieve massive things in life all alone. Therefore instead of only thinking about your HOW to get to your goals alone; think about WHO can help you achieve your goals more efficiently

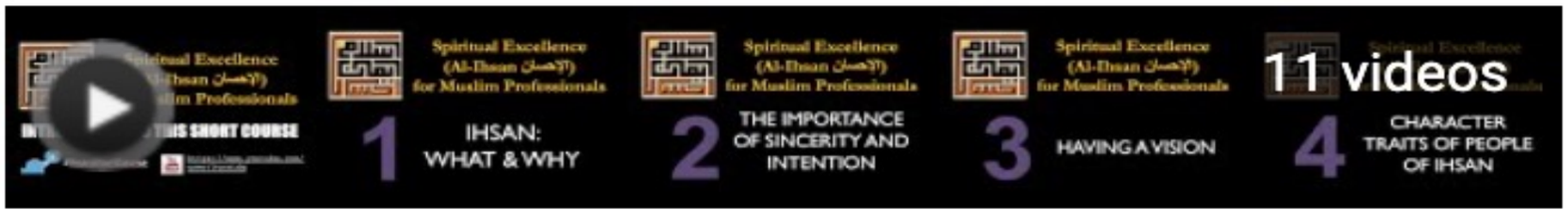


*It is better to be **independent** than **dependent** .. But the most effective people leverage **interdependence** through synergy and win-win partnerships*



Ethics; **Excellence**; Wisdom

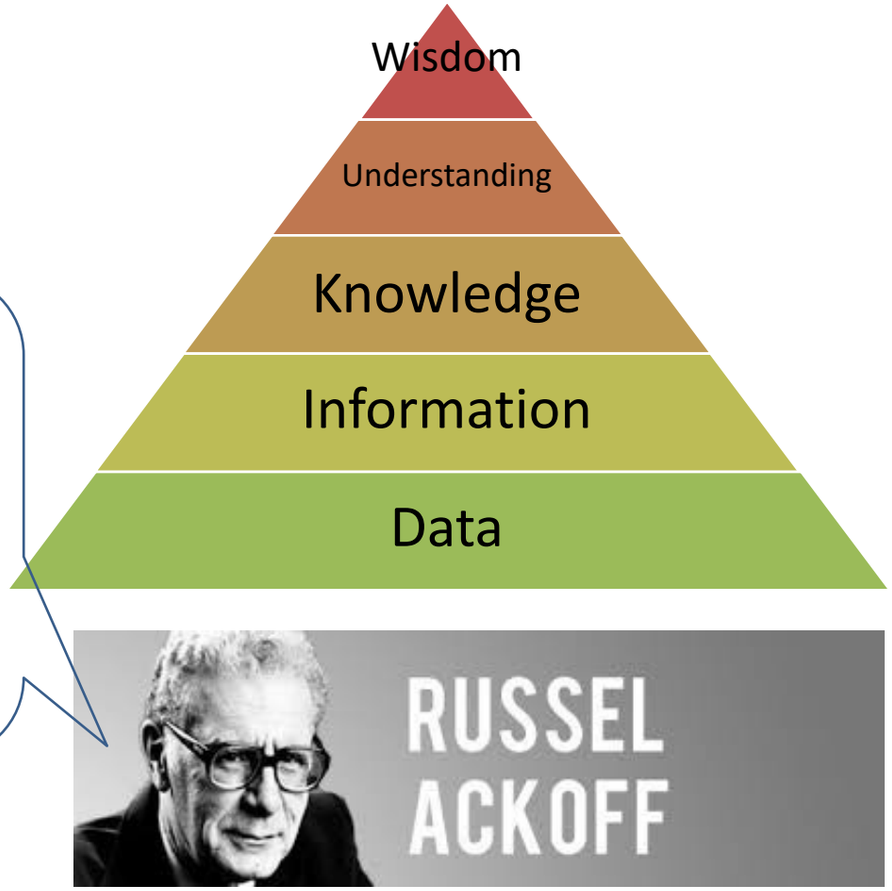




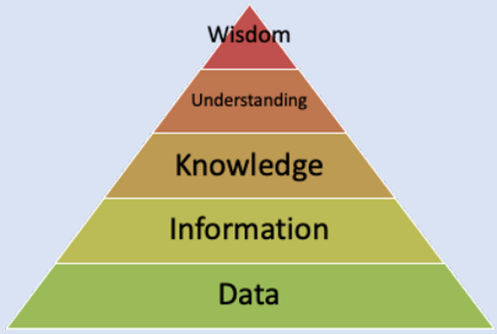
IHSAN for Muslim Professionals - A Short Course

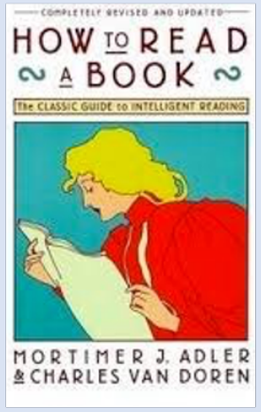
https://www.youtube.com/junaidq

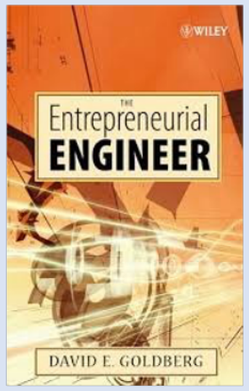
An ounce of information is worth a pound of data.
An ounce of knowledge is worth a pound of information.
An ounce of understanding is worth a pound of knowledge.
An ounce of wisdom is worth a pound of understanding.

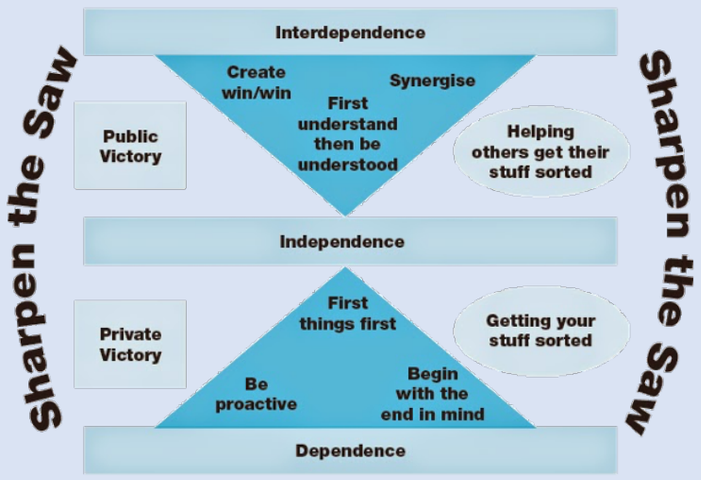


Questions/ Discussion Mini-Break









Engineering Education, Moving into 2020s:
Essential Competencies for Effective 21st Century Electrical & Computer Engineers

Junaid Qadir[1], Kok-Lim Alvin Yau[2], Muhammad Ali Imran[3], Ala Al-Fuqaha[4]
[1] Information Technology University, Lahore, Pakistan
[2] Sunway University, Malaysia
[3] University of Glasgow, UK
[4] Hamad bin Khalifa University, Qatar; Western Michigan University, USA

Dollars Versus Diplomas

What every student should know: Seven learning impediments and their remedies

Junaid Qadir

Many well-intentioned students fail to thrive in a university environment, not due to lack of effort, but because of a deficient mind-set and/or methodology. While a wide variety of problems can result in suboptimal learning performance, there are seven main categories of learning impediments that hinder student learning with alarming regularity. You may be facing one, or more, of these impediments. Not to fret though; each of these debilitating impediments can be surmounted with self-awareness and some effort, paving the way for enjoyable, and productive, learning experiences. In this article, I will identify the offending learning impediments and propose plausible remedial policies to counter each.

Is there anyone so wise as to learn by the experience of others?

—Voltaire

First impediment: Having a fixed mind-set

Psychologists have highlighted the self-enforcing propensity of the human mind-set in that human optimism and pessimism are both self-fulfilling. As Henry Ford said, "Whether you think you can or think you can't—you are right." A *fixed mind-set*, a mind-set model proposed by researcher Carol Dweck, takes the pessimistic approach to the concept of intelligence and skill development. It characterizes a belief that intelligence or talent are *fixed* human traits that cannot be significantly developed beyond a certain degree. Students with fixed mind-sets obsess over documenting their intelligence instead of focusing on developing their intelligence. Many fixed mind-set students are gifted students who have not learned how to struggle or the value of hard work. While they aced school subjects with little effort, when confronted with more significant intellectual challenges in a university setting that require more effort, these students capitulate easily while resorting to the excuse of a lack of aptitude. Why else would they—being such intelligent students—be confused?

What can be done: The antidote to this problem is to: 1) have a growth mind-set, 2) aim for mastery, and 3) use intrinsic motivation.

Catch phrase for the solutions: The sky is not the limit—your mind is. Conquer your mind to realize your potential.

Remedy 1: Have a growth mind-set

A growth mind-set student believes that hard work, grit, and

Careers and Grad Ed

Learning 101: The untaught basics

Junaid Qadir and Muhammad Ali Imran

Despite the accessibility of a wealth of information in the current era—books, universities, and online massive open online courses (MOOCs)—well-intentioned and hard-working students often fail to learn effectively due to deficient learning techniques or improper mind-sets. Two things, in particular, hinder students from achieving their potential. First, the students' intuition regarding how learning works is often flawed and counterproductive; second, despite significant progress in the research discipline of "learning sciences," these hard-earned scientific insights have not yet filtered their way through the research community to the students who stand to benefit most from this knowledge.

In this article, we aim to popularize the important insights from learning-science researchers by making it accessible to students, who continue to largely use suboptimal intuitive learning techniques (not knowing any better). The benefits of becoming wise to these insights are immediate and substantial: we can use this knowledge to understand and calibrate our learning, and it can also facilitate efficient learning.

Common learning impediments

To learn optimally, we need to discover how to learn effectively and also to identify and avoid common learning impediments. In our previous work, we identified seven common learning impediments (shown in Table 1). These impediments emerge from deficient mind-set (numbers 1 and 4), focus (2 and 3), support (5 and 6), and metacognition (7). These issues are not insurmountable; three potential remedies for each learning impediment are identified in our previous work as shown in Table 1. A major impediment—arguably the mother impediment of all the listed impediments, if considered broadly—is number 7, "Not learning how to learn." The overarching remedy for this problem is to spend time developing better metacognitive skills. Our aim is to succinctly present the relevant research findings in an accessible manner so that you can use this knowledge to improve your metacognitive skills and learning strategies.

Learning how to learn

The most useful piece of learning for the uses of life is to unlearn what is untrue.

—Antisthenes

Despite our intimacy with the task of learning (through the trials and errors of everyday living), research has shown that our metacognitive judgment about learning (i.e., our intuitive understanding of how we learn) is flawed, and many of our well-established and commonly accepted intuitive notions are not only inconsistent of scientific findings about learning but oppose the optimal course of action. Unfortunately, the common societal attitude, and the general assumption in colleges and universities, is that students can automatically figure out the art and science of learning by themselves. This is reflected in the way we emphasize the teaching of content and subjects in educational institutions but not the dissemination of information regarding an optimal learning strategy.

There is a Taoist saying that states, "To gain knowledge, one should add something every day, but to gain wisdom, one should subtract something every day." We propose that students should supplant the erroneous intuitive learning model with an improved metacognitive model of how learning

Thank you for your attention.

Questions/ comments/ feedback

Junaid.qadir@itu.edu.pk